

General Instructions

INSTALLATION INSTRUCTIONS

IMPORTANT SAFETY INFORMATION

Please read all instructions before servicing this equipment. Pay attention to all safety warnings and any other special notes highlighted in the manual. Safety markings are used frequently throughout this manual to designate a degree or level of seriousness and should not be ignored. **WARNING** indicates a potentially hazardous situation that if not avoided, could result in personal injury or death. **CAUTION** indicates a potentially hazardous situation that if not avoided, may result in minor or moderate injury or property damage.



WARNING:

ELECTRICAL SHOCK, FIRE OR EXPLOSION HAZARD

Failure to follow safety warnings exactly could result in serious injury or property damage.

Improper servicing could result in dangerous operation, serious injury, death or property damage.

Before servicing, disconnect all electrical power to furnace.

- Before servicing, disconnect all electrical power to the air handler.
- When servicing controls, label all wires prior to disconnecting. Reconnect wires correctly.
- Verify proper operation after servicing.

PROPOSITION 65 WARNING:

WARNING: This product contains chemicals known to the state of California to cause cancer.

WARNING: This product contains chemicals known to the state of California to cause birth defects or other reproductive harm.



WARNING:

Improper installation, service, adjustment, or maintenance may cause explosion, fire, electrical shock or other hazardous conditions which may result in personal injury or property damage. Unless otherwise noted in these instructions, only factory authorized kits or accessories may be used with this product.

This unit uses refrigerant R-454B. DO NOT use any other refrigerant in this unit. Ensure that any outdoor section is suitable for use with R-454B. DO NOT mix refrigerants.

Auxiliary devices which may be a potential ignition source shall not be installed in the duct work. Examples of such potential ignition sources are hot surfaces with a temperature exceeding 700°C and only electric switching devices which are sealed components, or intrinsically safe for use in flammable environments may be used. In addition to electric duct heaters, all other electrical accessory equipment within the ductwork of the unit should be inspected as well. Zoning systems, electro-static air filtration units and heating units should have particular attention applied to them.



**Refrigerant
Safety Group
A2L**



WARNING:

This appliance is not intended for use by persons (including children) with reduced physical, sensory or mental capabilities, or lack of experience and knowledge, unless they have been given supervision or instruction concerning use of the appliance by a person responsible for their safety. Children should be supervised to ensure that they do not play with the appliance.

This leak detector sensor (LDS) is only compatible with R-454B refrigerant. The R-454B refrigerant is a mildly flammable refrigerant. Because of this, there are requirements, including a minimum size, for the room in which the indoor portion of this system shall be installed within. Read the installation instructions for additional details.

When choosing an installation location, the contractor/installer must ensure that the equipment nameplate and markings continues to be visible and legible after its installation is completed. Markings and signs that are illegible shall be corrected.

Ensure the indoor units are designed, and factory configured to be installed on a fully ducted system and is Leak Detection Sensor (LDS) compatible.

Only auxiliary devices approved by Nordyne or that are declared suitable with R-454B shall be installed within the connecting ductwork.

The area shall be checked with an appropriate refrigerant detector prior to and during work, to ensure the technician is aware of potentially toxic or flammable atmospheres. Ensure that the leak detection equipment being used is suitable for use with all applicable refrigerants, i.e. non-sparking, adequately sealed or intrinsically safe.

The following checks shall be applied to installations using **FLAMMABLE REFRIGERANTS**:

- the actual refrigerant charge is in accordance with the room size within which the refrigerant containing parts are installed.
- the ventilation machinery and outlets are operating adequately and are not obstructed.
- marking on the equipment continues to be visible and legible. Markings and signs that are illegible shall be corrected
- refrigerating pipe or components are installed in a position where they are unlikely to be exposed to any substance which may corrode refrigerant containing components, unless the components are constructed of materials which are inherently resistant to being corroded or are suitably protected against being so corroded.

Leak detection sensors shall only be replaced with sensors specified by Nordyne.

Check that cabling will not be subject to wear, corrosion, excessive pressure, vibration, sharp edges or any other adverse environmental effects. The check shall also take into account the effects of aging or continual vibration from sources such as compressors or fans.

- All electrical wiring must be completed in accordance with local, state and national codes and regulations and with the National Electric Code (ANSI/NFPA 70) or in Canada the Canadian Electric Code Part 1 CSA C.22.1.

Locating the Leak Detection Sensor (LDS):

For Nordyne manufactured C75, C85 and air handle evaporator coils the LDS is located on the front delta plate of the coil.

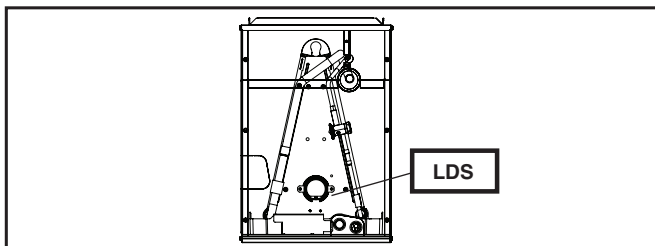


Figure 1. Location of the LDS in indoor coil delta plate.

For Nordyne manufactured small package units the LDS is located on a bracket installed at the foot of the blower housing immediately behind the blower service access door.

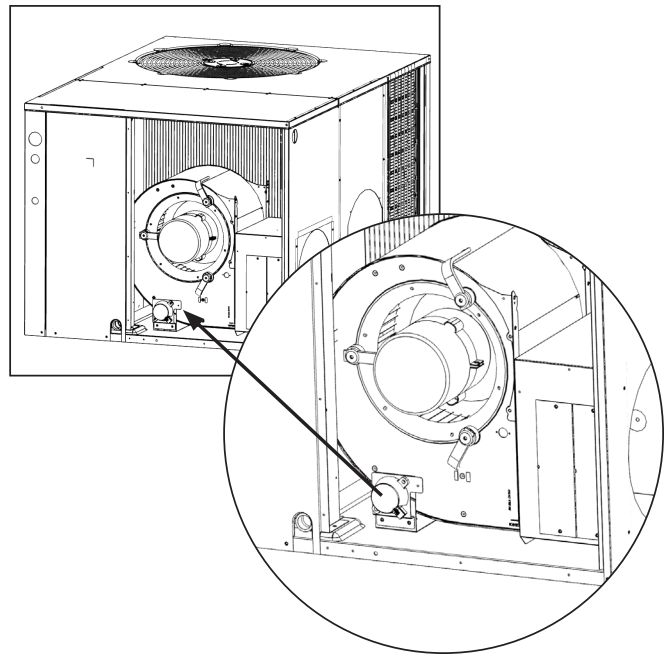


Figure 2. Location of LDS in small package unit.

INSTALLATION INSTRUCTIONS:

If this kit is being used for replacement of damaged or cracked sensor only, verify the mating connector is not damaged and proceed to replace the damaged sensor with the new sensor provided in the kit. Reconnect the sensor cable and reroute the cable in the same manner as it was originally installed.

If this kit is being used to replace not only a damaged or cracked sensor but the mating connector as well:

1. Cut the six wires that feed into the mating connector and strip fresh ends from the sensor cable with appropriate lengths.
2. Reattach the 6" pig tails to the sensor cable with the wire nuts that have been provided in the kit, ensuring to match the wire colors.
3. Reconnect the sensor cable to the LDS.

Important note regarding sensor cable routing:LDS

The mating connector used on the sensor cable is not designed to be water tight. Water intrusion in the connector may cause a short or cause the contacts to corrode thereby causing a LDS failure. If the LDS is being installed in a vertical orientation, the mating connector must always face downwards and the sensor cable must be routed in such a way that creates a J-loop to drain any condensation that forms on the cable away from the mating connector. If the sensor is being installed on a horizontal surface, the mating connector must not point directly into the airstream to prevent any blow off condensation from entering the connector. Wire the sensor as per the wiring diagram of the furnace. An example wiring layout is shown on [Page 3](#).

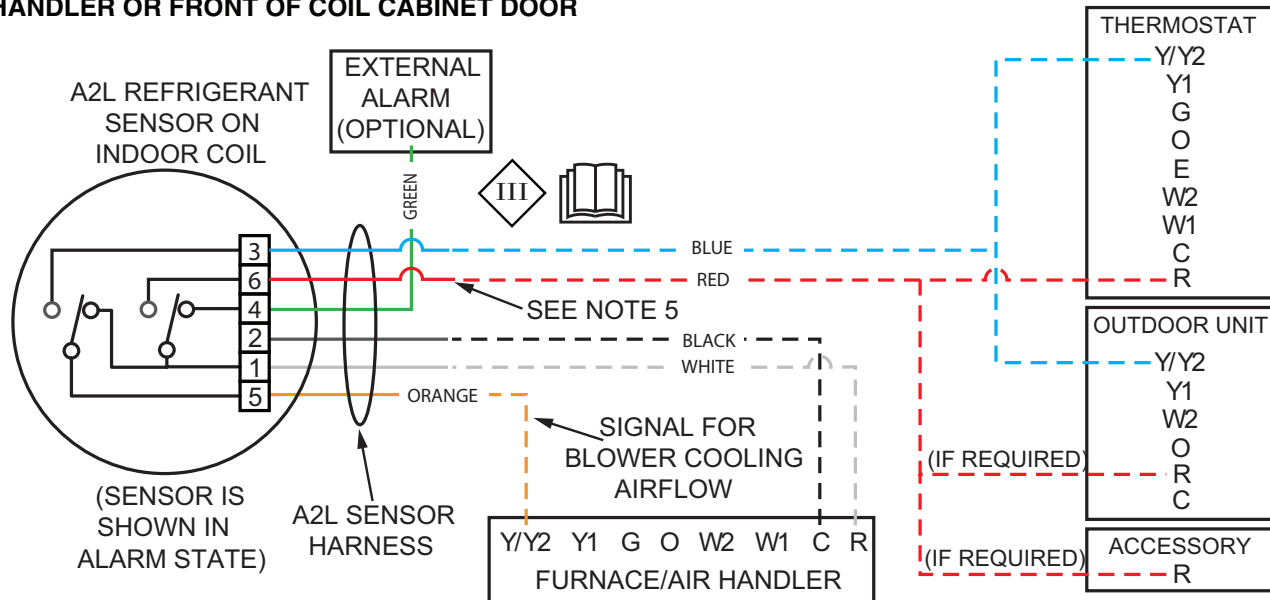
If two or more sensors need to be installed in the application, then they must be wired in series as shown on [Page 4](#).

If the application requires an Auxiliary A2L relay, Nordyne kit, PN # 1049245 provides a relay and pigtail harness.

WIRING DIAGRAM

A2L SENSOR FIELD WIRING DIAGRAM

NOTE TO INSTALLER! APPLY THIS LABEL NEXT TO EXISTING WIRING DIAGRAM IN FURNACE / AIR HANDLER OR FRONT OF COIL CABINET DOOR



NOTES:

1. Disconnect all power before servicing.
2. The transformer may have a dual voltage primary tap. Match the tap position with the supply voltage used.
3. Connect 24VAC/40VA Class 2 circuit to A2L Refrigerant Sensor and thermostat.
4. If the internal wiring is replaced, use only 105°C copper wire of the same gauge.
5. All 24VAC power for external devices (i.e. "R") is to be powered from A2L Refrigerant Sensor 'Pin6' output.
6. All other low voltage wiring not shown in this diagram to be per standard thermostat wiring. See installation instructions for example wiring.

LEGEND:

FIELD WIRING	----
LOW VOLTAGE	_____
HIGH VOLTAGE	—————

1. Coupez l'alimentation avant de faire l'entretien.
2. Le transformateur peut avoir un robinet principal à double tension. Agencez la position du robinet au type de tension de l'installation.
3. Branchez un circuit 24 V c.a. / 40 VA Classe 2 au capteur de frigorigène A2L et au thermostat.
4. Si le câblage interne est remplacé, utilisez seulement un fil de cuivre 105 °C du même calibre.
5. Toute alimentation 24 V c.a. pour les appareils externes (soit « R ») doit être acheminée depuis la sortie « tige 6 » du capteur de frigorigène A2L.
6. Tous les autres fils basse tension non illustrés dans ce schéma doivent être conformes au câblage standard du thermostat. Voir les installations d'instruction pour des exemples de câblage.



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Figure 3. A2L Sensor Field Wiring Diagram.

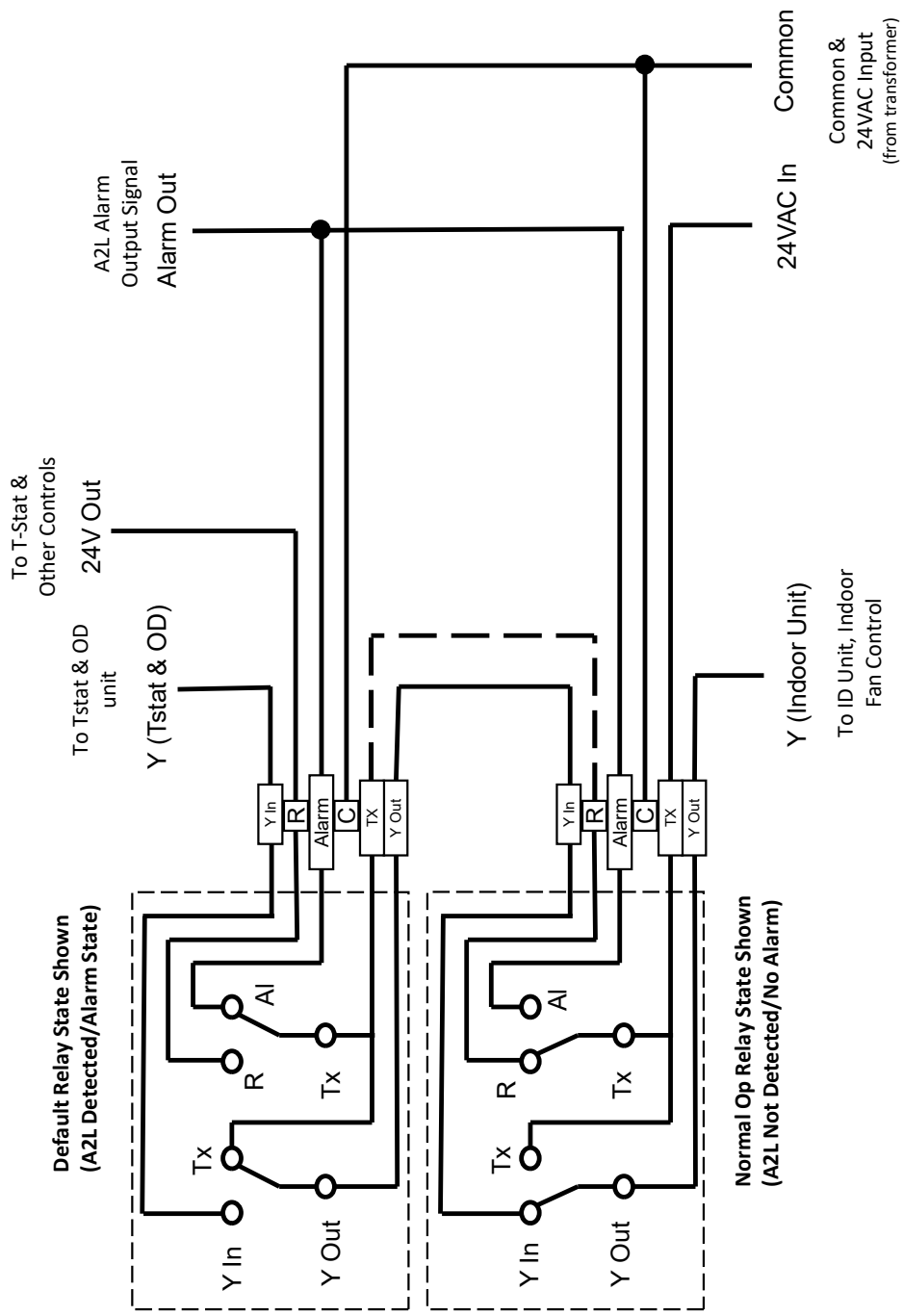


Figure 4. Series Sensor Wiring for systems with 2 or more sensors.

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